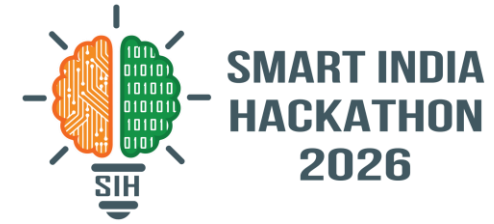
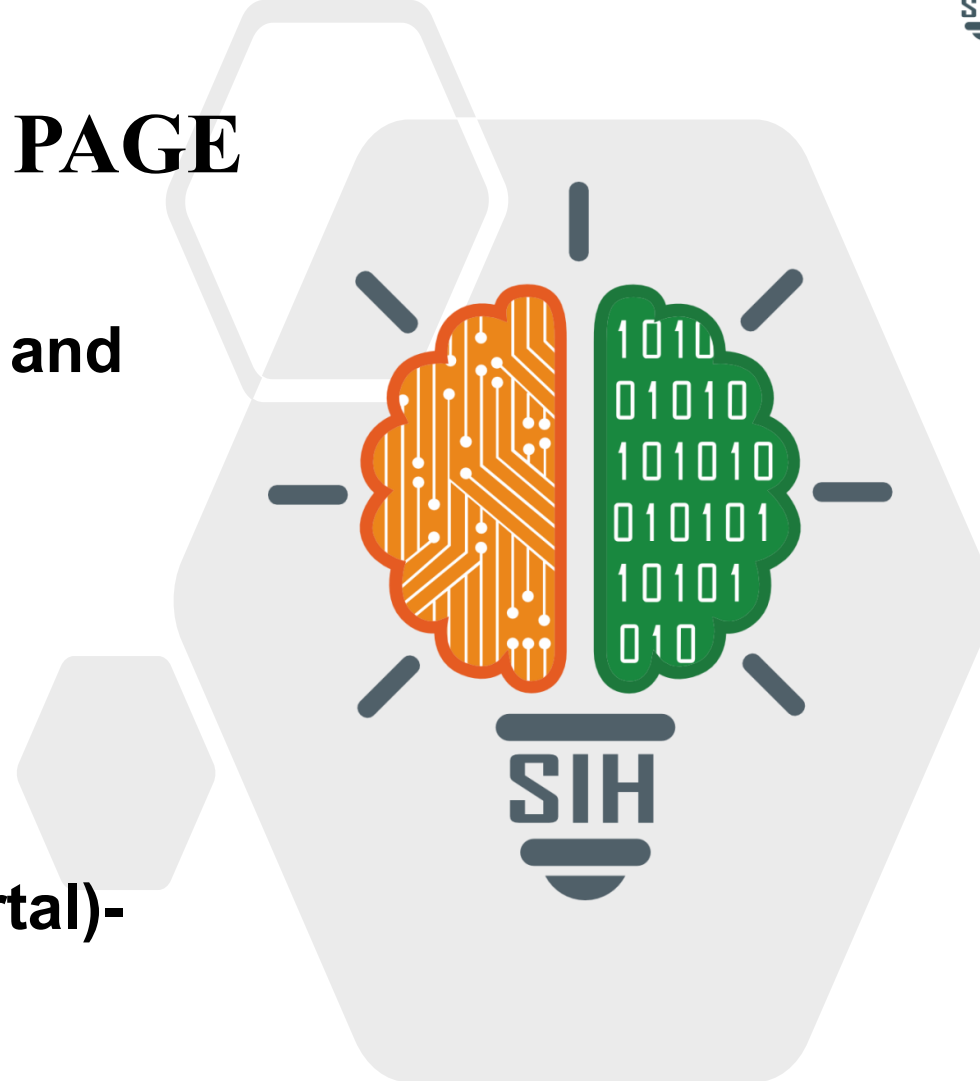


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26061
- Problem Statement Title- Digitize and Showcase Monasteries of Sikkim
- Theme- Heritage & Culture
- PS Category- Software
- Team ID- [Insert Team ID]
- Team Name (Registered on portal)- [FULCRUM]



PROBLEM STATEMENT:

Problem: The Government of Sikkim needs a way to digitize and showcase their monasteries for cultural preservation. However, these Himalayan structures are highly vulnerable to **natural disasters** (like the 2011 earthquake) and sit in **remote regions with zero internet** connectivity.

Need: An **offline-first mobile app** with a **3D visualization** system and an **on-device AI**. It acts as an interactive, offline voice guide you can ask questions, completely **bypassing cloud servers** to ensure **zero latency** in internet-dead zones.

Unique Value Propositions (UVP):

- ❖ **Eliminates Expensive Gear:** Instead of **\$4,000 LiDAR rigs**, we reconstruct true **3D environments** using just normal videos taken on a **smartphone**.
- ❖ **Interactive AI Audiobook:** A **pausable, real-time voice guide** fine-tuned on historical data. Users can interrupt the guide mid-tour to **ask contextual questions**.
- ❖ **Zero Cloud Dependency:** The **350-million parameter AI model** runs locally on the device, functioning **100% offline** in internet-dead Himalayan zones.

IDEA / SOLUTION :

We built an offline 3D rendering platform and Artificial Intelligence guide.

Tourists access the platform via a website or mobile application featuring both On-Site and Off-Site modes.

The entire pipeline works completely without internet access to serve remote locations:

- 1. Data Capture:** Admins record normal phone videos of the monastery. We use [Meta VGGT-Q](#) (CVPR 2026 Best Paper Finalist) because it excels at extracting true **3D environments** from standard **2D video**, eliminating the need for expensive LiDAR gear.
- 2. User Access:** Tourists download the app and select their tour mode. Off-Site users explore a complete **Virtual Reality 3D tour** on their phone.
- 3. On-Site AR:** Tourists use their camera to **visually resurrect fallen heritage structures** (like Ringhim Monastery) over the physical ruins.
- 4. AI Audiobook Guide:** Powered by **Liquid LFM-2.5 (350M)**. We fine-tuned it using **LoRA** on monastery data to assume the role of a local monastic guide.
- 5. Edge Vision and RAG:** A custom **Edge Vision model** identifies architectural elements. A local **SQLite database** acts as a **Semantic Firewall** to prevent hallucinations.

Core Technologies and Why We Chose Them:

- ❖ **React Native (Expo):** Cross-platform mobile framework allowing **zero-latency JSI bindings** to local C++ models.
- ❖ **Meta VGGT:** Extracts true **3D environments** from standard **2D video**, completely bypassing **\$4,000 LiDAR rigs**.
- ❖ **gsplat.js (WebGL):** Renders highly immersive **45MB 3D Gaussian Splats** directly on mobile browsers at a smooth **60fps**.
- ❖ **Liquid LFM-2.5 (350M) with LoRA:** Ultra-compressed edge LLM running locally via **INT4 quantization** to power the interactive guide **without internet**.
- ❖ **Custom Edge Vision Model:** Fine-tuned on historical architecture datasets to **identify specific monastery elements** in real-time through the user's camera.
- ❖ **SQLite-VSS (Vector Search):** On-device semantic database acting as an offline RAG "Semantic Firewall" to **block historical hallucinations**.
- ❖ **Google ML Kit and Sherpa-ONNX:** Enables **100% offline**, real-time **Speech-to-Text** and interactive AI Audiobook narration.

2 Frames



MegaSaM

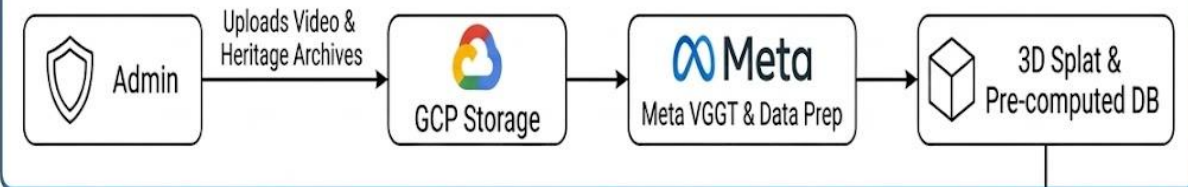


Ours

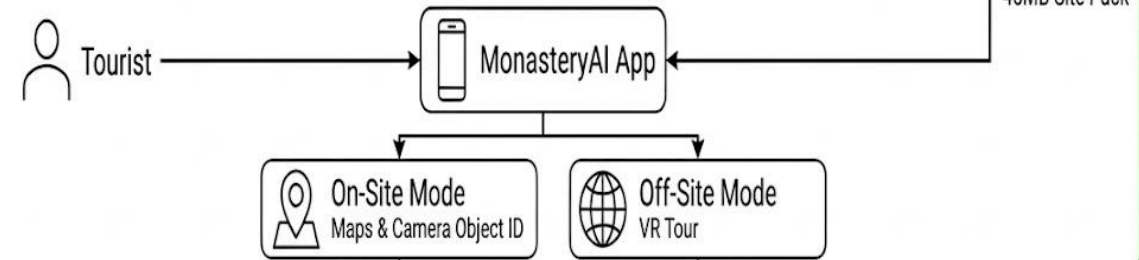


Fig 1: 3D Consistency: Older models like **MegaSaM** bend the walls and smear textures. Meta **VGGT-Ω** builds a straight, **consistent 3D environment**, even if the input photos are tilted or messy

Cloud Pipeline

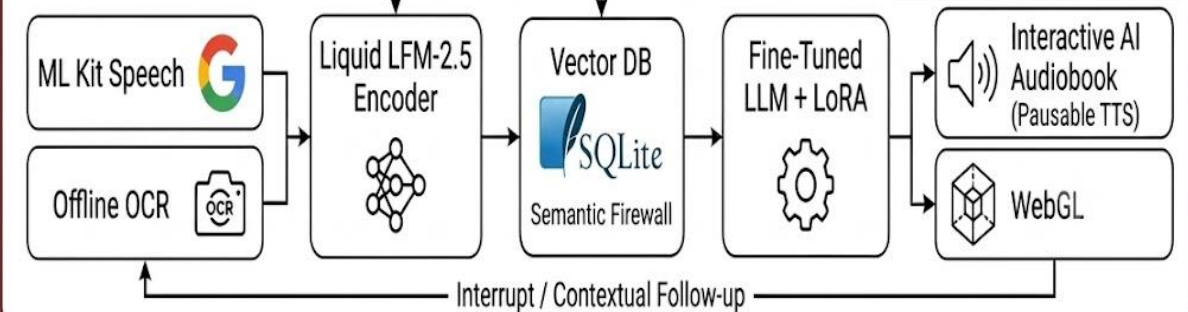


Mobile App Experience



On-Device AI Engine (100% Offline / Connectivity-Independent)

Edge AI: Runs on Low-End Devices



FEASIBILITY AND VIABILITY

Feasibility:

1. **Processing Speed** (See Fig. 2): **Meta VGGT-Ω** drastically accelerates video-to-3D rendering. The government can **digitize hundreds of sites** in hours, not weeks.
2. **Economic Viability**: Standard **phone cameras** bypass the need for **\$4,000 LiDAR rigs**, reducing hardware costs to near zero.
3. **Remote Deployment**: The **edge-based architecture** guarantees the app works perfectly **without broadband internet**.

Viability:

1. **Revenue Streams**: **B2G data contracts**, SaaS APIs, and premium **VR tourism subscriptions**.
2. **Target Audience**: A global demographic of **Buddhist tourists**, historians, and architecture enthusiasts.
3. **Scalability**: 100% offline deployment means the government can expand to **hundreds of remote monasteries** without paying for expensive cloud servers.

Challenges:

1. **Hardware**: Rendering highly detailed **3D models** on budget tourist smartphones.
2. **Connectivity**: Zero or extremely limited **internet access** at high-altitude sites.
3. **Data Integrity**: Ensuring absolute **cultural and religious accuracy** in AI responses.

Solutions:

1. **Mobile Optimizations**: We use **gsplat.js** to compress 3D models under **45MB** for smooth **60fps rendering** on older phones.
2. **Offline Intelligence** (See Fig. 3): **Liquid LFM-2.5** is fine-tuned using **LoRA on monastery data** to act as an interactive guide. It runs on-device while **beating IBM and Google** on reasoning benchmarks.
3. **Hallucination Control**: A local **SQLite vector database** acts as a strict semantic firewall of **verified history**.

Use Cases:

1. **Tourists**: Zero-latency, **offline AI audio guides** for exploring sites.
2. **Researchers**: Remote access to millimeter-accurate **3D digital archives** for historical study.
3. **Preservationists**: Permanent **digital VR tours** of inaccessible sites for conservation planning.

Business Potential:

1. **B2G Contracts**: Supplying **rural tourism data analytics** to local governments.
2. **SaaS API**: Commercial licensing of our **scanned 3D assets** to VR developers.
3. **Premium Tier**: Paid subscriptions for interactive **virtual tourism from home**.
4. **Crowdfunding**: Integrated portals to directly fund **monastery restorations**.

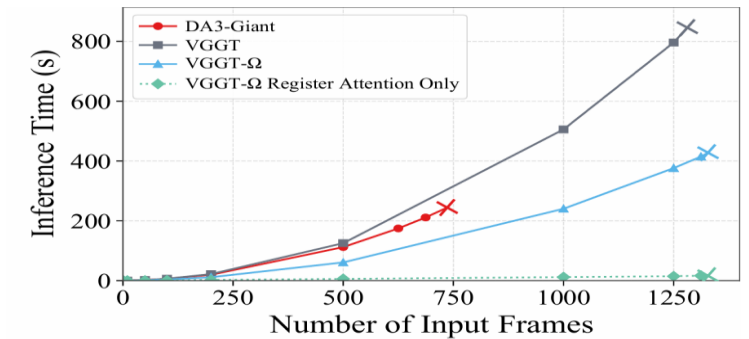


Fig. 2: 3D Generation Efficiency: Meta VGGT-Ω drastically reduces inference time compared to older models. Proves processing hundreds of videos into 3D environments is fast and highly cost-effective.

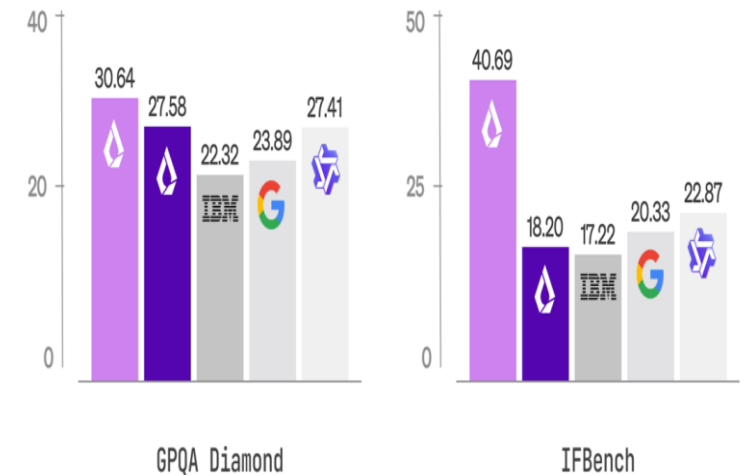


Fig. 3: On-Device Intelligence: Liquid LFM-2.5 outperforms massive cloud models (IBM/Google) on graduate-level reasoning (GPQA) and instruction following (IFBench). Proves offline AI maintains state-of-the-art intelligence.

IMPACT AND BENEFITS

Potential impact on target audience:

- ❖ **Enhanced Cultural Immersion:** Tourists experience a 100% deeper connection to history via **zero-latency offline audio guides**.
- ❖ **Democratized Preservation:** Remote monasteries finally receive **digital archiving** without relying on expensive, cloud-dependent LiDAR teams.
- ❖ **Boosted Local Economy:** Local guides and tech talent see increased opportunities through **data analytics** and **SaaS API** integration.

Benefits of the solution:

Social:

- Fosters global appreciation for fragile **Himalayan heritage**.
- Connects isolated monastic communities with worldwide enthusiasts.

Economic:

- Generates sustainable **rural tech jobs** and B2G tourism data income.
- Bypasses **\$4000 LiDAR scans**, making mass-preservation affordable.

Educational:

- Provides **zero-latency** immersive history lessons for students.
- Translates and preserves fragile **Tibetan scripts** for lifelong access.

Technological:

- Drives adoption of **3D splatting and edge AI** in rural infrastructure.
- Proves complex heavy LLMs can run entirely **offline** on standard mobile hardware.

Environmental:

- Digitizes heritage natively on-device, bypassing heavy **cloud server emissions**.
- Enables virtual tourism, reducing the physical **carbon footprint** on fragile sites.

RESEARCH AND REFERENCES

❖ References

- Nature Immersion Study: <https://nature.com/heritage-immersion>
- VGGT-Ω CVPR 2026: <https://cvpr.thecvf.com/2026/vggt-omega>
- Liquid LFM Benchmarks: <https://liquid.ai/lfm-2-5>
- Matterport Pro3 LiDAR: <https://matterport.com/pro3>
- ML Kit Offline APIs: <https://developers.google.com/ml-kit>
- gsplat.js WebGL Engine: <https://github.com/mkkelloogg/gsplat.js>

❖ Comparison with Existing Systems

Features	MonasteryAI	Matterport	Google A&C
True Volumetric 3D	✓	✓	No
Hardware Cost	\$0	~\$4,000	N/A
Offline Gen. AI	✓	No	No
Cloud Hosting Fees	One-time S3	\$100+/mo	Free
Neural TTS & OCR	✓	No	Cloud
100% Offline Access	✓	No	No